

Enterprise Email Triage via QLoRA Fine-Tuning of a 3B SLM

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1 · Problem & Domain

The problem. Operations teams triage a firehose of internal email by hand: what is this asking for, how urgent is it, who owns the follow-up, does it need to be escalated. I turn a single cleaned email body into strict, machine-actionable JSON against a fixed schema, emitting `null` / empty when a field is genuinely absent rather than guessing:

```
{
  "intent": "request | status_update | scheduling | approval_request | escalation | fyi | other",
  "urgency": "low | medium | high",
  "requires_response": true,
  "action_items": [ {"owner": "string|null", "task": "string", "deadline": "string|null"} ],
  "escalation_flag": false
}
```

Data source. The public **Enron email corpus** (FERC release; Kaggle [wcukierski/enron-email-dataset](#), ~517k messages). A load-bearing design choice: **owner and deadline, when non-null, must be verbatim substrings of the email**. That turns hallucination into a free mechanical check (`span in body`) and forces grounding rather than plausible invention. `task` may be paraphrased.

Why this is a real AI opportunity, and why Track B (fine-tuning) fits. Triage is high-volume, repetitive, and structured — the shape that pays back automation — but the hard part is *behavioural*: rigid schema adherence, a consistent 7-way taxonomy, and calibrated abstention. Also helps in keeping corporate emails on-prem vs exposing to 3P API.

2 · Approach & Algorithm Decisions

Fine-tuning vs. prompting / few-shot / RAG / zero-shot

I ran the zero-shot base model through the identical harness first, so this is measured, not asserted.

Prompting alone leaves the base Qwen2.5-3B at **86% schema validity** and **26% intent accuracy** — it drifts across the taxonomy.

- **RAG is wrong by construction.** There is no external corpus to retrieve — the email *is* the whole context.
- **Few-shot** burns context window per call and still does not lock output structure or taxonomy at 3B scale.

Structured-output tooling with schema validation would constrain the *space* of outputs; it cannot improve the *choice* within that space. That semantic judgment requires **weight adaptation**.

Base model: Qwen2.5-3B-Instruct

Constraint: ≤13B, open-weight, must fit QLoRA on a free T4/P100. Rejected alternatives:

- **Mistral-7B-v0.3** — weaker structured-output priors; 7B QLoRA roughly triples epoch time on a T4, which would have cost me the ablation within the time budget.
- **Llama-3.2-3B-Instruct** — genuinely close. Chose Qwen2.5-3B for stronger out-of-the-box JSON adherence, which raises the *baseline* floor and makes the fine-tuning delta a cleaner read.

QLoRA vs. full fine-tuning vs. other PEFT

Full fine-tuning of 3B needs ~40 GB+ in optimizer states alone; a T4 has 16 GB — infeasible. **QLoRA** (4-bit NF4 base + double quant, fp16 LoRA adapters) fits comfortably. Among PEFT methods, **IA³** has far fewer trainable params but less capacity to learn a 7-way taxonomy; **prompt-tuning** adds soft tokens without touching the weights that carry the semantic mapping — both rejected on capacity, not format.

Rank / alpha / targets and the ablation

$r=16$ balances capacity against overfit at ~360 training examples; $\alpha=32$ is the conventional 2x scaling; $\text{dropout}=0.05$. I target attention + MLP because MLP layers carry format/style adaptation. I tested that hypothesis directly with an **attention-only** ablation at identical r , seed, data, epochs, LR, and batch.

Full config: 2 epochs · LR 2e-4 cosine, warmup 0.03 · effective batch 16 (bs 2 × grad-accum 8) · paged_adamw_8bit · fp16 · max_seq 1280 · seed 42.

Data pipeline & the teacher-label

From 517k emails: exact-dedup and trigram based near-dedup → ~204k clean pool → sample ~360 training (targeted for 800 samples but stopped early due to rate limit) + 50 held-out messages. Labels do not exist, so a teacher LLM (**Gemini-3.5-Flash-Lite**) synthesises them; the 50-example test set is then **hand-audited into a gold set** (needed minor corrections in 4-5 examples only).

3 · Results & Error Analysis

All metrics on the 50-example hand-audited gold set. Base is zero-shot through the identical harness and generation budget. Classification metrics for the base are scored only on its parseable outputs.

Metric	Base (0-shot)	Tuned — attn+MLP	Tuned — attn-only
Schema validity	86%	100%	100%
Intent accuracy (7-way)	26%	70%	64%
Action-item count acc.	63%	72%	72%
requires_response acc.	77%	86%	78%
Urgency accuracy (3-way)	79%	68%	74%
Abstention recall	88%	78%	85%
Unsupported-span rate	0% (15 spans)	7% (2/27)	4% (1/25)
escalation_flag acc.	98%	98%	98%

Clear Improvements: Schema validity 86 → 100 and intent 26 → 70 are the real capability lift — the model learned to always emit valid JSON and to internalise the taxonomy, neither of which prompting delivered.

Grey Area: Fine-tuning made the model a more *active extractor*: it predicts many more action-item spans (15 → 27), gets the count right more often (63 → 72), but abstains slightly less (88 → 78/85) and occasionally grabs a non-verbatim span (0 → 4–7%). Base Qwen is **conservative** — it abstains well and rarely hallucinates a span, but cannot produce valid JSON or the taxonomy; the tuned model is **assertive** — it locks format and learns intent, at a small precision/grounding cost.

Where it fails, and why.

- Urgency did not improve** (79 → 68/74). Urgency is subjective; could be due to *teacher-label noise* not a model regression.
- Unsupported spans rose** (0 → 4–7%). The assertive shift trades a little grounding precision for recall.
- escalation_flag is uninformative.** 98% is majority-class dominated (escalations are rare in the gold set); probably F1 could be a better metric here.

Ablation. No clean winner: attn+MLP leads intent (+6) and requires_response (+8); attn-only leads urgency (+6) and abstention (+7) with fewer unsupported spans at **~1/4 the trainable params** (~7.4M vs ~30M). Both hit 100% schema. The honest call is **attn-only is roughly as good for a quarter of the cost**. A larger eval set could help with significance of these small differences in performance.

Learning vs. memorization. Every number above is on emails the model **never saw in training** — the 50 gold messages are sampled disjoint from the ~360 training messages and hand-audited, so the gains are generalisation. Overfitting pressure is low by design (2 epochs, ~1% of params, dropout 0.05). The stronger control I would add next is an **entity-perturbation test** — swap names/dates/companies in the gold set and

confirm scores hold; a memoriser would collapse on grounding, a learner would not. I expect it to hold, since `owner / deadline` are learned as a copy-from-context operation, not memorised entities.

3a · Hallucination risk — baseline vs. fine-tuned

The grounding rule the hallucination detector: a non-null `owner / deadline` must be a verbatim substring of the source email, so `unsupported_span_rate` is a deterministic hallucination measure.

Model	owner/deadline spans	Ungrounded (fabricated)	Rate
Base (zero-shot)	25	0	0.0%
Tuned — attn+MLP	27	2	7.4%
Tuned — attn-only	25	1	4.0%

Fine-tuning introduced a hallucination risk the base model did not have. The base fabricated zero spans — because it is conservative and predicts few spans at all. The tuned model extracts more aggressively (25 → 27 spans) and, in doing so, occasionally asserts an `owner / deadline` that is not literally in the email. This is the precision cost of the recall-leaning shift, and it is exactly where a fine-tuned extractor is most dangerous: a fabricated owner or deadline is a *confident, plausible, wrong* operational instruction.

Mitigation The grounding rule is enforced at the serving boundary: any ungrounded `owner / deadline` is coerced to `null` (the model abstains on that field rather than asserting a fabrication). `task` is grounding-exempt by design and never altered. This drives the unsupported-span rate to an **enforced 0%** with a single auditable rule — a concrete, zero-model structured-output guard rather than a hope that the weights behave.

3b · Catastrophic forgetting probe

I probed 15 general MCQs (MMLU-style knowledge + reasoning) and 10 open-ended instruction-following prompts, scored base vs. tuned through the same harness:

	Base	Tuned — attn+MLP	Tuned — attn-only
General MCQ accuracy	86.7% (13/15)	86.7% (13/15)	86.7% (13/15)
JSON-bleed rate (open-ended)	0/10	0/10	0/10
Instruction-following	coherent	coherent	coherent

No measurable regression. General MCQ accuracy is unchanged, and critically the **JSON-bleed rate is 0%** — asked for a haiku, the tuned model still writes a haiku, not triage JSON. Both ablation variants behave identically.

4 · Production & Limitations

Production consideration — a validation boundary. Serve the adapter merged into a 4-bit 3B on a single T4 with batched generation. Because the schema is a hard contract, the format layer should be **grammar-constrained decoding** (guaranteed-valid JSON and enums on the first pass, no wasted retries), and the content layer the **grounding guard** (null any `owner / deadline` that is not a verbatim span). Together these convert the residual 7% unsupported-span rate into a guaranteed-valid, guaranteed-grounded external contract — the fine-tuned weights supply the *semantics*, structured-output tooling supplies the *guarantees*.

Limitation to address before real deployment. Performance is **ceilinged by silver-label quality** — the model can be at most as good as the teacher, and the flat urgency metric is that ceiling showing through on a subjective field. The next step is a modest set of **human-labelled in-house emails** to fine-tune on the domain-real distribution and re-anchor the subjective fields, plus multi-seed evaluation on a larger gold set to turn the ablation sub-deltas and the forgetting/hallucination bounds into real findings.